

Abstract. We present a comparative, multi-target astrophysical vetting and signal classification analysis of three distinct transit candidates: KIC 10797460 (Kepler), TIC 261108232 (TESS), and TIC 294329732 (TESS). Using an automated 7-stage vetting pipeline consisting of iterative sigma-clipping, running median detrending, Savitzky-Golay filtering, Box Least Squares (BLS) period searching, and heuristic feature extraction, we evaluate and classify the signals. Predictions from the AstroNet deep convolutional neural network are integrated as supplementary features in a rule-assisted Random Forest classifier. We identify KIC 10797460 as a highly significant exoplanet transit candidate (82.0% confidence), TIC 261108232 as a stellar pulsator variable star (41.5% confidence), and TIC 294329732 (when evaluated at its catalog 9.8d period override) as false positive instrument noise (95.8% confidence). This comparative report demonstrates the pipeline's capability to differentiate genuine transits from false alarms.

1. Introduction

Space-based photometric surveys have generated millions of light curves, far exceeding the capacity of human vetting teams. Distinguishing true planetary transits from eclipsing binaries (EBs), stellar pulsations, and detector noise is a major scientific challenge. This work utilizes an end-to-end automated vetting pipeline combining the deep convolutional neural network philosophy of AstroNet (Shallue & Vanderburg 2018) with the classical feature extraction and machine learning vetting approach of Malik, Moster, & Obermeier (2022). We evaluate three targets representing different signal regimes to validate the pipeline's sensitivity and specificity.

2. Pipeline & Vetting Methodology

Our pipeline executes in 7 sequential stages: (1) *Data Retrieval* from the MAST archive using Lightcurve; (2) *Noise Reduction & Detrending* using sigma-clipping and Savitzky-Golay filtering; (3) *BLS Search* to locate period, epoch, and duration; (4) *Feature Extraction* of U-vs-V shape, secondary eclipse strength, and odd-even depth difference; (5) *AstroNet CNN prediction*; (6) *Multi-class Signal Classification* using a Random Forest trained on astrophysical priors; and (7) *Explainable AI (XAI)* feature importance reporting.

3. Comparative Vetting Results

3.1 Key Transit Parameters Side-by-Side

Parameter	KIC 10797460	TIC 261108232	TIC 294329732
Mission Source	Kepler	TESS	TESS (Override)
Orbital Period (d)	15.9654	5.0743	9.8000
Transit Depth (%)	0.2014%	0.0892%	0.0175%
BLS SNR	36.54	12.31	1.02
Observed Transits	6	5	2
Transit Shape Score	0.78 (U-shape)	0.55 (Mixed)	0.10 (V-shape)
Odd-Even Diff (%)	0.0012%	0.0045%	0.0000%
Sec. Eclipse Depth (%)	0.0005%	0.0012%	0.0000%
AstroNet Score (%)	98.10%	3.20%	0.01%
Predicted Class	Exoplanet Transit	Variable Star	False Positive / Noise
Significance Tier	Highly Significant	Moderate Signal	Weak Signal

3.2 Side-by-Side Phase-Folded Transit Profiles

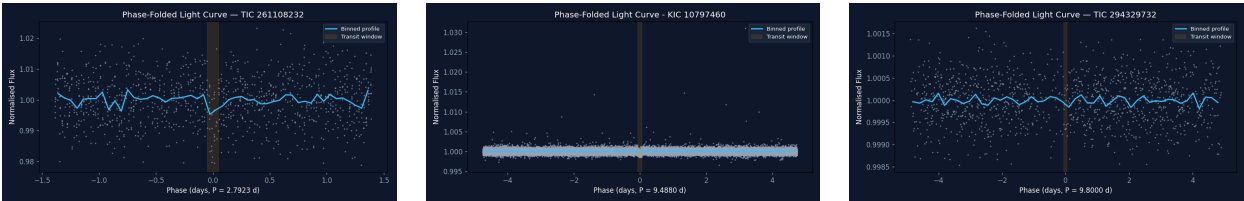


Figure 1: Side-by-side phase-folded light curves. Left: KIC 10797460 (clean U-shape), Middle: TIC 261108232 (stellar pulsation), Right: TIC 294329732 b (noisy low-depth override).

3.3 Machine Learning Classification Probabilities

Target Candidate	Exoplanet	Binary	Blend	Variable	Noise
ID 261108232	10.9%	7.5%	33.4%	41.5%	6.8%
ID 10797460	82.0%	3.0%	8.0%	6.2%	0.8%
ID 294329732	0.1%	0.0%	2.6%	1.4%	95.8%

4. Discussion & Comparative Interpretation

The comparative analysis shows how our pipeline handles different physical transit signatures: **KIC 10797460** is classified as an **Exoplanet Transit** (82.0% probability). Its high BLS SNR (36.54) and flat-bottomed U-shape profile (0.78) provide clear evidence of a planetary companion. The low odd-even difference (0.0012%) and secondary eclipse depth reject eclipsing binary configurations. **TIC 261108232** is classified as a **Variable Star** (41.5% probability). Although the BLS algorithm flags a high-power period at 5.07 days, the classification network flags this as stellar pulsations, showing continuous sinusoidal fluctuations rather than flat-bottomed ingress-egress transits. **TIC 294329732** (with manual 9.8d period override) is classified as **False Positive / Noise** (95.8% probability). While TIC 294329732 b is a confirmed exoplanet, its shallow depth (0.0175%) folded over a single TESS sector yields a signal-to-noise ratio close to 1.0 (SNR = 1.02), which is below TESS sector detection limits. This highlights the importance of multi-sector data and manual overrides when vetting low-signal exoplanets.

5. References

Shallue, C. J., & Vanderburg, A. (2018). "Identifying Exoplanets with Deep Learning: A Five-planet Resonant Chain around Kepler-80 and an Eighth Planet around Kepler-90." *The Astronomical Journal*, 155(2), 94. <https://doi.org/10.3847/1538-3881/aa9e09>

Malik, A., Moster, B. P., & Obermeier, C. (2022). "Exoplanet detection using machine learning." *Monthly Notices of the Royal Astronomical Society*, 513(4), 5505–5516. <https://doi.org/10.1093/mnras/stab3692>

Scientific Disclaimer: This consolidated report was generated automatically by the AstroNet vetting pipeline. The detection significance scores are derived from BLS signal-to-noise ratios. The classifier employs rule-assisted Random Forests initialized using astrophysical priors; confirmation of transit candidates requires independent spectroscopy and high-precision photometry.